
Combinatorial development of nebulized mRNA delivery formulations for the lungs

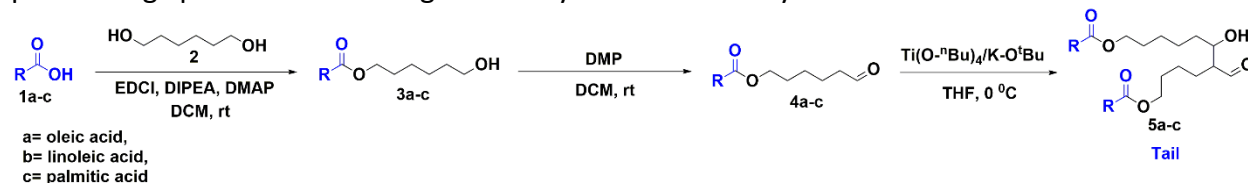
In the format provided by the
authors and unedited

Table of Contents

1. Synthesis of Lipids
2. Supplementary Figures
 - a. Supplementary Figure 1. Characterizing effects of nebulization on LNPs
 - b. Supplementary Figure 2. *In vivo* nebulized mRNA delivery with systemic LNPs
 - c. Supplementary Figure 3. DLS of DOE LNPs
 - d. Supplementary Figure 4. Characterization of lead T1-5 LNP from DOE screen
 - e. Supplementary Figure 5. Reproducibility of nebulization *in vitro*
 - f. Supplementary Figure 6. Characterization of bPEG20K effects on nebulized LNP formulation and mouse lungs
 - g. Supplementary Figure 7. Comparison of whole-body exposure and nose-only exposure systems for *in vivo* nebulized mRNA-LNP delivery
 - h. Supplementary Figure 8. Characterization of NaAc dialyzed LNPs
 - i. Supplementary Figure 9. Effects of nebulization on mRNA integrity
 - j. Supplementary Figure 10. Comparison between purified and unpurified ionizable lipids
 - k. Supplementary Figure 11. A549 cell screening of novel ionizable lipids
 - l. Supplementary Figure 12. Trans-epithelial electrical resistance (TEER) of air-liquid interface (ALI) cultures
 - m. Supplementary Figure 13. ALI culture screen of lead lipid tail variants
 - n. Supplementary Figure 14. ALI culture screen of lead lipid headgroup variants
 - o. Supplementary Figure 15. Biodistribution of luciferase in lung lobes
 - p. Supplementary Figure 16. Quantification of FFL protein expression in homogenized lung tissues
 - q. Supplementary Figure 17. Optimized nebulization formulation for reductive amination lipids
 - r. Supplementary Figure 18. Physicochemical characterization of optimized nebulization formulation for IR-117-17 and IR-19-Py LNPs
 - s. Supplementary Figure 19. Degradation analysis of IR-117-17
 - t. Supplementary Figure 20. Safety and toxicity of nebulized LNP
 - u. Supplementary Figure 21. Kinetics of luciferase expression in the lung of *Scnn1b*-Tg mice
 - v. Supplementary Figure 22. Ai14 mouse model transfection analysis
 - w. Supplementary Figure 23. Transfection of alveolar spaces
3. Supplementary Table 1. Formulation parameters for DOE study in Figure 1

Synthesis of Lipids

All solvents and reagents were obtained commercially and used as such unless noted otherwise. ^1H and ^{13}C NMR spectra were recorded in CDCl_3 at room temperature using a Bruker Ultrashield 400 MHz instrument. Spectra were processed with Chemical shifts reported as parts per million (ppm) relative to TMS (0.00) for ^1H . Silica gel chromatography was performed on ISCO CombiFlash Rf+ Lumen Instruments using ISCO RediSep Rf Gold Flash Cartridges (particle size: 20-40 microns). All final compounds were confirmed by mass spectrometry using direct injection method on QTOF-HRMS (Agilent) coupled to an Agilent Infinity 1260 LC system. Only top performing lipids after screening were fully characterized by ^1H and ^{13}C NMR.



Scheme 1: synthesis of tails used in reductive amination library synthesis.

Synthesis of 6-hydroxyhexyl oleate, **3a**:

Oleic acid (5g, 17.7 mmol), *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide hydrochloride (EDC HCl, 5.1g, 26.6 mmol), hexane-1,6-diol (10.5g, 88.5 mmol), 4-(dimethylamino) pyridine (DMAP, 1.1g, 8.9 mmol) and *N,N*-diisopropylethylamine (DIPEA, 6.2 ml, 35.4 mmol) were dissolved in dichloromethane (200 ml). The reaction was stirred at room temperature under nitrogen for 18 h, then washed with a saturated aqueous sodium bicarbonate solution. The organic layer was separated, washed with brine, dried over Na_2SO_4 , filtered, and the filtrate was evaporated under vacuum. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-hydroxyhexyl oleate, **3a** (5.8g, 15.2 mmol, 86%) as a colorless oil. ^1H NMR (400 MHz, CDCl_3) δ 5.43 – 5.29 (m, 2H), 4.09 (t, J = 6.7 Hz, 2H), 3.67 (t, J = 6.5 Hz, 2H), 2.31 (t, J = 7.5 Hz, 2H), 2.03 (q, J = 6.5 Hz, 4H), 1.73 – 1.53 (m, 6H), 1.49 – 1.17 (m, 24H), 0.95 – 0.83 (m, 3H).

Synthesis of 6-oxohexyl oleate, **4a**:

6-hydroxyhexyl oleate, **3a** (5.8g, 15.2 mmol) was dissolved in dichloromethane (300 ml) followed by Dess-Martin Periodinane (9.6g, 22.8 mmol). The reaction was stirred under nitrogen at room temperature for 2 hr. After confirmation of reaction completion by thin layer chromatography, sodium thiosulfate pentahydrate (50% w/v, 200 ml) was added to the reaction and left to stir for additional 15 mins. Organic layer was thereafter separated, washed with brine, dried over Na_2SO_4 , filtered, and the filtrate was evaporated under vacuum. The residue obtained was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-oxohexyl oleate, **4a** (3.5g, 9.2 mmol, 61%) as a colorless oil. ^1H NMR (400 MHz, CDCl_3) δ 9.79 (s, J = 1.8 Hz, 1H), 5.42 – 5.33 (m, 2H), 4.09 (t, J = 6.6 Hz, 2H), 2.47 (td, J = 7.3, 1.7 Hz, 2H), 2.31 (t, J = 7.5 Hz, 2H), 2.03 (q, J = 6.5 Hz, 4H), 1.67 (dp, J = 15.1, 7.8 Hz, 7H), 1.48 – 1.21 (m, 23H), 0.90 (t, J = 6.8 Hz, 3H).

Synthesis of 5-formyl-6-hydroxyundecane-1,11-diyl dioleate, **5a**:

Titanium (IV) butoxide (3.1g, 9.2 mmol) and potassium tert-butoxide (1.0g, 9.2 mmol) were dissolved in anhydrous tetrahydrofuran (200ml) and left to stir under nitrogen for 5 mins at room temperature. The mixture was then cooled to 0 °C. Thereafter, (6-oxohexyl oleate, **11** (3.5g, 9.2 mmol) was added to the reaction and stirring was continued for another 45 mins. The reaction was quenched with 1N HCl (100 ml), and then extracted with ethyl acetate (250 ml). Organic layer

was separated, dried over Na₂SO₄, filtered, and the filtrate was evaporated under vacuum. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 5-formyl-6-hydroxyundecane-1,11-diyl dioleate, **5a** (1.7g, 2.3 mmol, 25%) as a brown oil. ¹H NMR (400 MHz, CDCl₃) δ 9.79 (dd, *J* = 7.1, 2.3 Hz, 1H), 5.45 – 5.32 (m, 4H), 4.09 (td, *J* = 6.6, 3.3 Hz, 4H), 2.31 (t, *J* = 7.6 Hz, 4H), 2.05 (dq, *J* = 12.8, 6.5, 5.7 Hz, 7H), 1.87 (d, *J* = 5.3 Hz, 2H), 1.74 – 1.46 (m, 15H), 1.46 – 1.07 (m, 42H), 0.90 (t, *J* = 6.7 Hz, 6H).

Synthesis of 6-hydroxyhexyl (9Z,12Z)-octadeca-9,12-dienoate, **3b**

6-hydroxyhexyl (9Z,12Z)-octadeca-9,12-dienoate, **3b** was made using the same procedure for the synthesis of 6-hydroxyhexyl oleate, **3a** above. ¹H NMR (500 MHz, CDCl₃) δ 5.47 – 5.26 (m, 4H), 4.09 (t, *J* = 6.7 Hz, 2H), 3.67 (t, *J* = 6.5 Hz, 2H), 2.80 (t, *J* = 6.7 Hz, 2H), 2.31 (t, *J* = 7.6 Hz, 2H), 2.07 (q, *J* = 7.0 Hz, 4H), 1.74 – 1.53 (m, 7H), 1.49 – 1.18 (m, 18H), 0.91 (t, *J* = 6.8 Hz, 3H).

Synthesis of 6-oxohexyl (9Z,12Z)-octadeca-9,12-dienoate, **4b**

6-oxohexyl (9Z,12Z)-octadeca-9,12-dienoate, **4b** was made using the same procedure for the synthesis of 6-oxohexyl oleate, **4a** above. ¹H NMR (400 MHz, CDCl₃) δ 9.79 (s, 1H), 5.59 – 5.15 (m, 4H), 4.09 (t, *J* = 6.6 Hz, 2H), 2.79 (t, *J* = 6.4 Hz, 2H), 2.31 (t, *J* = 7.6 Hz, 2H), 2.06 (dd, *J* = 8.1, 5.7 Hz, 4H), 1.68 (dq, *J* = 14.4, 7.2 Hz, 8H), 1.44 – 1.22 (m, 16H), 0.91 (t, *J* = 6.7 Hz, 3H).

Synthesis of 5-formyl-6-hydroxyundecane-1,11-diyl (9Z,9'Z,12Z,12'Z)-bis(octadeca-9,12-dienoate), **5b**

5-formyl-6-hydroxyundecane-1,11-diyl (9Z,9'Z,12Z,12'Z)-bis(octadeca-9,12-dienoate), **5b** was made using the same procedure for the synthesis of 5-formyl-6-hydroxyundecane-1,11-diyl dioleate, **5a**. ¹H NMR (400 MHz, CDCl₃) δ 9.78 (dd, *J* = 7.0, 2.3 Hz, 1H), 5.47 – 5.24 (m, 8H), 4.09 (td, *J* = 6.6, 3.0 Hz, 4H), 3.86 (t, *J* = 6.1 Hz, 1H), 2.79 (t, *J* = 6.4 Hz, 4H), 2.31 (t, *J* = 7.5 Hz, 4H), 2.06 (dd, *J* = 7.9, 5.9 Hz, 9H), 1.74 – 1.60 (m, 10H), 1.48 – 1.16 (m, 36H), 0.91 (t, *J* = 6.8 Hz, 6H).

Synthesis of 6-hydroxyhexyl palmitate, **3c**

6-hydroxyhexyl palmitate **3c** (octadecanoic acid) was made using the same procedure for the synthesis of 6-hydroxyhexyl oleate, **3a** above. ¹H NMR (500 MHz, CDCl₃) δ 4.09 (t, *J* = 6.7 Hz, 2H), 3.67 (q, *J* = 6.2 Hz, 2H), 2.31 (t, *J* = 7.6 Hz, 2H), 1.64 (tp, *J* = 21.2, 6.7 Hz, 7H), 1.48 – 1.36 (m, *J* = 3.6, 3.0 Hz, 4H), 1.36 – 1.20 (m, 22H), 0.90 (t, *J* = 6.9 Hz, 3H).

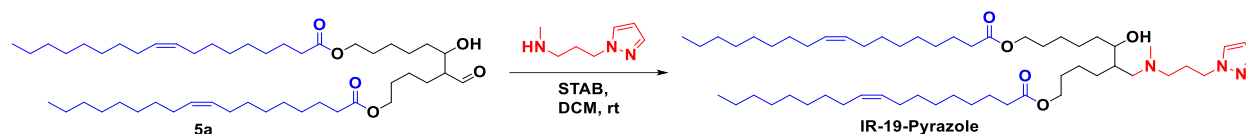
Synthesis of 6-oxohexyl palmitate, **4c**

6-oxohexyl palmitate **4c** was made using the same procedure for the synthesis of 6-oxohexyl oleate, **4a** above. ¹H NMR (500 MHz, CDCl₃) δ 9.79 (s, 1H), 4.09 (t, *J* = 6.6 Hz, 2H), 2.48 (td, *J* = 7.3, 1.7 Hz, 2H), 2.31 (t, *J* = 7.5 Hz, 2H), 1.75 – 1.53 (m, 7H), 1.28 (s, 24H), 0.90 (t, *J* = 6.9 Hz, 3H).

Synthesis of 5-formyl-6-hydroxyundecane-1,11-diyl dipalmitate, **5c**

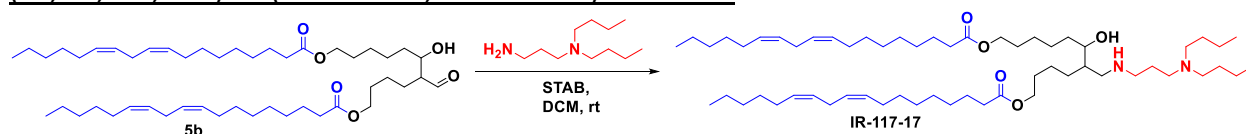
5-formyl-6-hydroxyundecane-1,11-diyl dipalmitate, **5c** was made using the same procedure for the synthesis of 5-formyl-6-hydroxyundecane-1,11-diyl dioleate, **5a**. ¹H NMR (500 MHz, CDCl₃) δ 9.79 (s, 1H), 4.09 (qd, *J* = 6.7, 5.8, 2.4 Hz, 4H), 3.91 – 3.78 (m, 1H), 2.31 (t, *J* = 7.6 Hz, 5H), 2.07 (d, *J* = 5.6 Hz, 2H), 1.91 – 1.75 (m, 3H), 1.75 – 1.49 (m, 16H), 1.46 – 1.21 (m, 43H), 0.90 (t, *J* = 6.9 Hz, 6H).

Synthesis of 5-(((3-(1H-pyrazol-1-yl)propyl)(methyl)amino)methyl)-6-hydroxyundecane-1,11-diyl dioleate, **IR-19-pyrazole**



N-methyl-3-(1H-pyrazol-1-yl)propan-1-amine (0.18g, 1.30 mmol) was added to a solution of 5-formyl-6-hydroxyundecane-1,11-diyl dioleate, **5a** (0.5g, 0.66 mmol) in dichloromethane (100 ml) and left to stir under nitrogen at room temperature for 1 hr. Sodium triacetoxymethylborohydride (0.42g, 1.97 mmol) was then added and the reaction was left to stir at room temperature overnight. Solvent was evaporated off, and the residue was purified by silica gel chromatography (10% methanol in dichloromethane with 0.1% NH₄OH) to give IR-19-Pyrazole as a colorless oil (0.37g, 0.42 mmol, 63%). QTOF MS (ESI): *m/z* calcd for C₅₅H₁₀₂N₃O₅⁺ (M+H), 884.78140; found, 884.7793. ¹H NMR (400 MHz, CDCl₃) δ 7.52 (d, *J* = 1.8 Hz, 1H), 7.45 (dd, *J* = 5.6, 2.3 Hz, 1H), 6.24 (t, *J* = 2.0 Hz, 1H), 5.42 – 5.31 (m, 4H), 4.24 – 4.12 (m, 2H), 4.07 (q, *J* = 7.1 Hz, 4H), 3.52 (t, *J* = 8.3 Hz, 1H), 2.76 – 2.38 (m, 3H), 2.37 – 2.18 (m, 8H), 2.17 – 1.92 (m, 11H), 1.62 (tdd, *J* = 14.3, 12.1, 10.1, 5.0 Hz, 10H), 1.48 – 1.20 (m, 46H), 1.14 – 1.00 (m, 1H), 0.94 – 0.83 (m, 6H). ¹³C NMR (101 MHz, CDCl₃) δ 173.98, 173.91, 173.89, 139.42, 130.02, 129.99, 129.76, 129.71, 105.22, 64.39, 63.97, 63.93, 63.59, 60.39, 59.87, 54.89, 49.52, 49.46, 41.92, 41.86, 39.41, 39.22, 35.24, 34.37, 34.33, 31.90, 29.76, 29.70, 29.52, 29.32, 29.30, 29.19, 29.15, 29.12, 29.04, 28.86, 28.81, 27.22, 27.17, 26.23, 26.20, 24.99, 22.68, 14.12.

Synthesis of 5-(((3-(dibutylamino) propyl) amino) methyl)-6-hydroxyundecane-1,11-diyl (9Z,9'Z,12Z,12'Z)-bis(octadeca-9,12-dienoate) **IR-117-17**



N, *N*'-dibutylpropane-1,3-diamine (0.1g, 0.54 mmol) was added to a solution of 5-formyl-6-hydroxyundecane-1,11-diyl (9Z,9'Z,12Z,12'Z)-bis(octadeca-9,12-dienoate), **5b** (0.21g, 0.28 mmol) in dichloromethane (50 ml) and left to stir under nitrogen at room temperature for 1 hr. Sodium triacetoxymethylborohydride (0.18g, 0.84 mmol) was then added and the reaction was left to stir at room temperature overnight. Solvent was evaporated off, and the residue was purified by silica gel chromatography (10% methanol in dichloromethane with 0.1% NH₄OH) to give 5-(((3-(dibutylamino) propyl) amino) methyl)-6-hydroxyundecane-1,11-diyl (9Z,9'Z,12Z,12'Z)-bis(octadeca-9,12-dienoate), **IR-117-17** as a light brown oil (0.16g, 0.17 mmol, 61%). QTOF MS (ESI): *m/z* calcd for C₅₉H₁₁₁N₂O₅⁺ (M+H), 927.84875; found, 927.8480. ¹H NMR (500 MHz, CDCl₃) δ 5.45 – 5.29 (m, 8H), 4.08 (t, *J* = 6.7 Hz, 4H), 3.75 (dt, *J* = 9.5, 2.9 Hz, 1H), 3.56 (td, *J* = 7.6, 2.8 Hz, 1H), 2.96 – 2.86 (m, 1H), 2.79 (t, *J* = 6.7 Hz, 4H), 2.76 – 2.59 (m, 3H), 2.55 – 2.36 (m, 6H), 2.31 (td, *J* = 7.6, 2.2 Hz, 4H), 2.07 (q, *J* = 7.0 Hz, 8H), 1.80 – 1.58 (m, 12H), 1.51 – 1.15 (m, 46H), 0.93 (dq, *J* = 11.3, 8.0, 7.0 Hz, 12H). ¹³C NMR (126 MHz, CDCl₃) δ 173.97, 173.91, 130.23, 130.22, 130.07, 130.03, 128.07, 128.04, 127.92, 127.90, 75.95, 64.41, 64.39, 64.05, 64.03, 53.76, 53.73, 52.84, 52.78, 49.54, 49.30, 42.00, 41.80, 36.11, 34.38, 34.34, 32.90, 31.53, 31.51, 31.50, 29.65, 29.62, 29.35, 29.21, 29.20, 29.16, 29.13, 29.08, 29.04, 28.97, 28.81, 28.79, 27.24, 27.21, 26.81, 26.36, 26.22, 26.20, 25.67, 25.66, 25.64, 25.34, 25.00, 24.12, 23.54, 22.58, 20.72, 14.11, 14.08, 14.06.

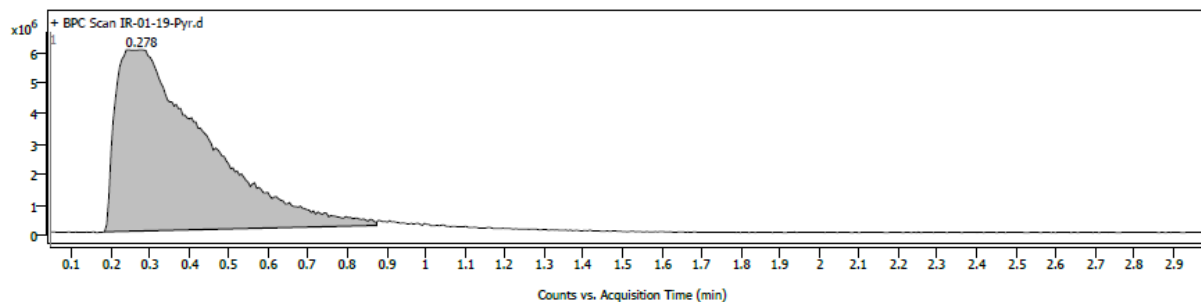
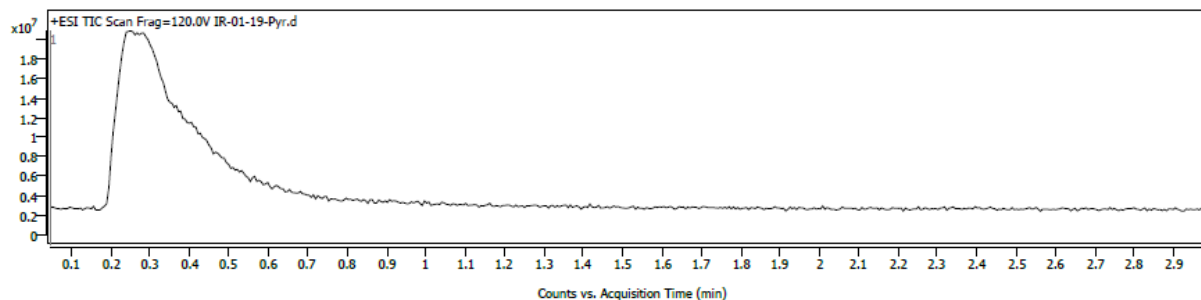
Analysis Report



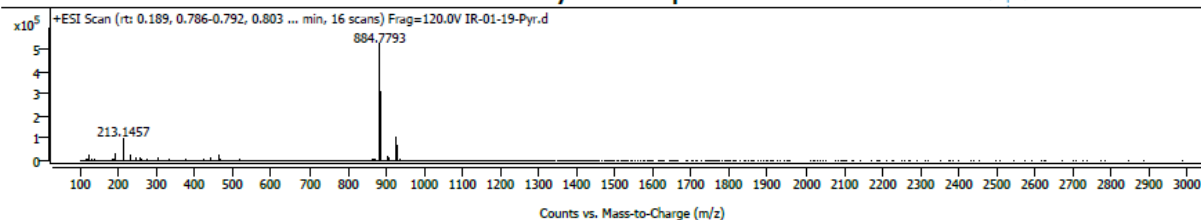
Sample Information

Name	19-pyr	Data File Path	D:\MassHunter\Data\Users\Anderson\AEiraj\Repeat117_4CR\IR-01-19-Pyr.d
Sample ID		Acq. Time (Local)	3/2/2022 12:20:10 PM (UTC-04:00)
Instrument	QTOF	Method Path (Acq)	D:\MassHunter\Methods\Users\Anderson\DI_04182019.m
MS Type	QTOF	Version (Acq SW)	6200 series TOF/6500 series Q-TOF 10.1 (48.0)
Inj. Vol. (ul)	1	IRM Status	All ions missed
Position	P1-A3	Method Path (DA)	D:\MassHunter\Data\Users\Anderson\AEiraj\Repeat117_4CR\IR-01-19-Pyr.d\AcqData\DI_04182019.m
Plate Pos.		Target Source Path	D:\MassHunter\PCDL\default.csv
Operator		Result Summary	1 qualified (1 targets)

Sample Chromatograms

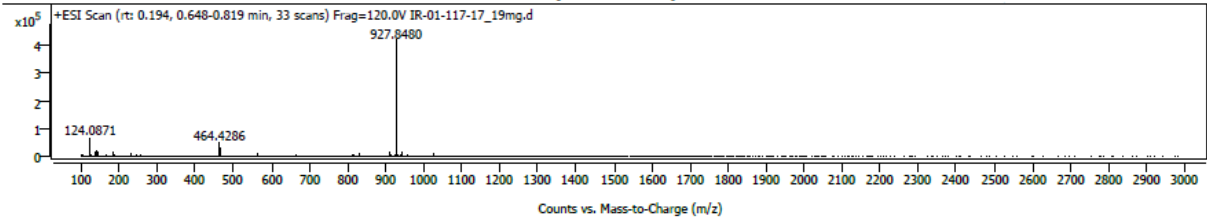
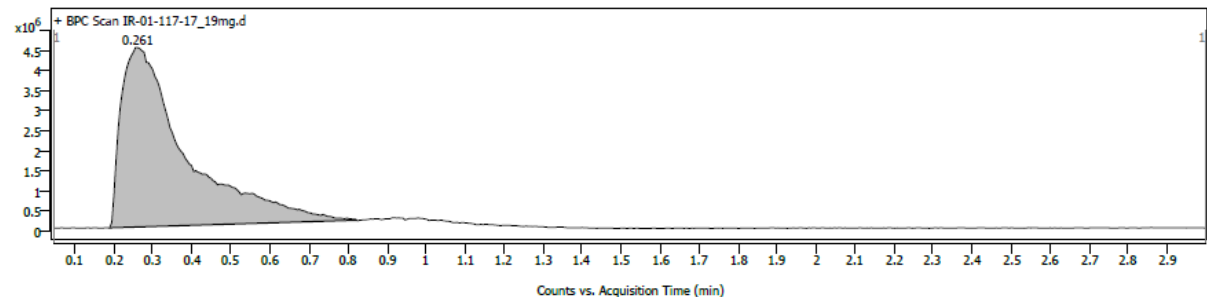
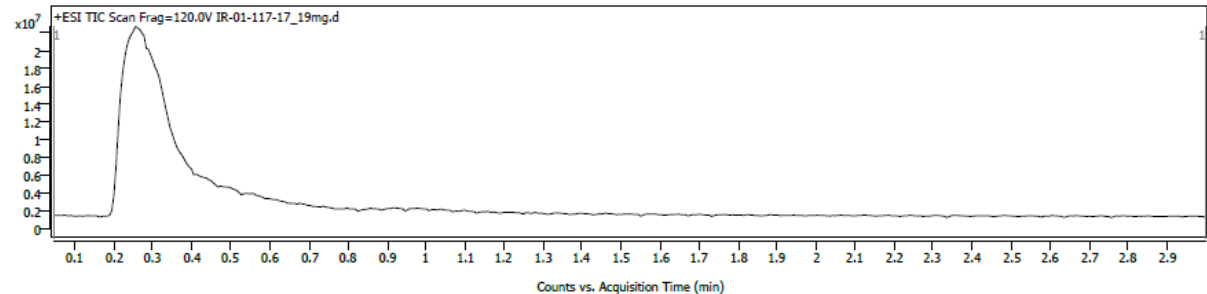


Analysis Report

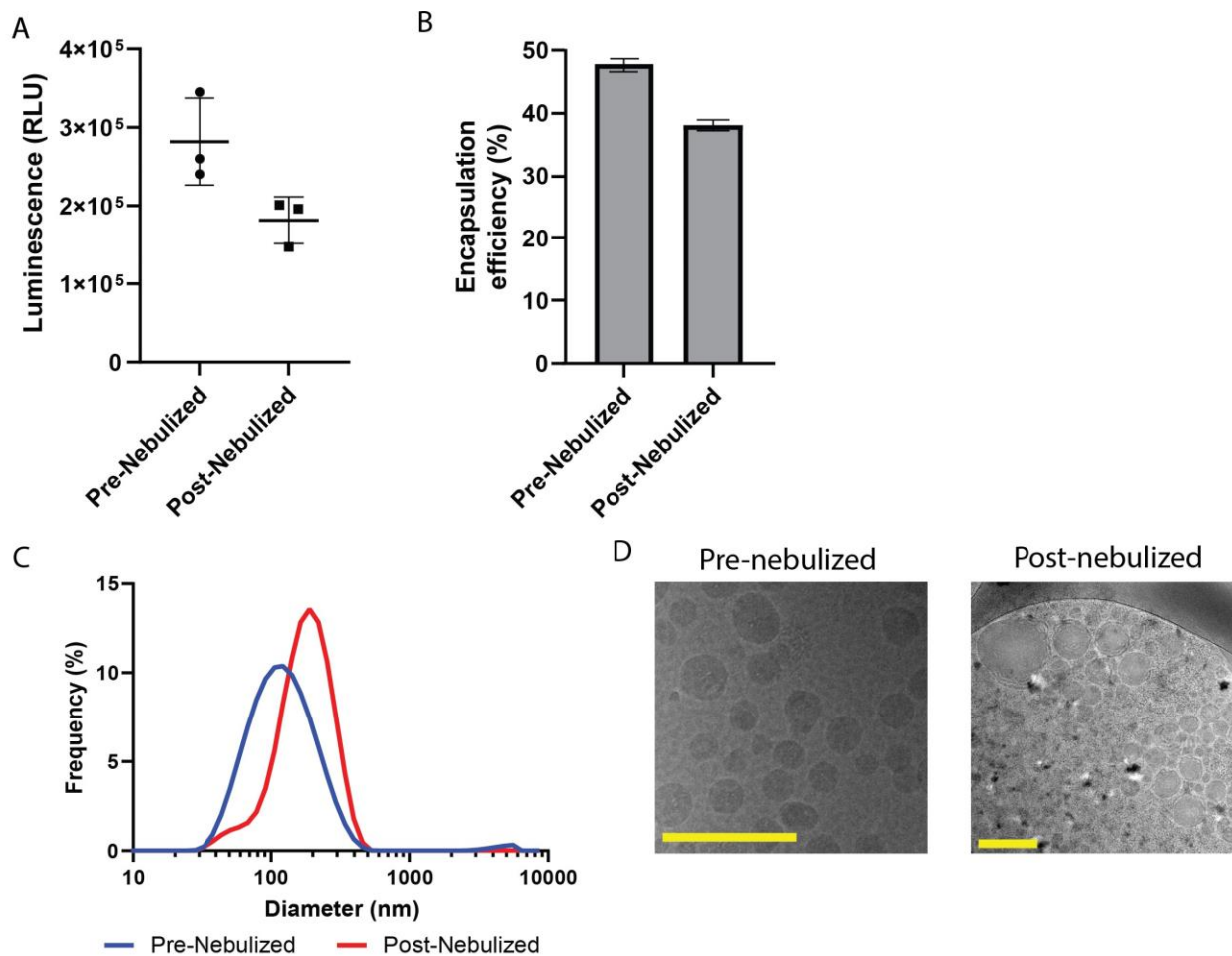


Analysis Report			
 Agilent Trusted Answers			
Sample Information			
Name	IR-117-17_19mg	Data File Path	D:\MassHunter\Data\Users\Anderson\AEiraj>Last\IR-01-117-17_19mg.d
Sample ID		Acq. Time (Local)	3/30/2023 5:57:49 PM (UTC-04:00)
Instrument	QTOF	Method Path (Acq)	D:\MassHunter\Methods\Users\Anderson\DI_04182019.m
MS Type	QTOF	Version (Acq SW)	6200 series TOF/6500 series Q-TOF 10.1 (48.0)
Inj. Vol. (ul)	1	IRM Status	All ions missed
Position	P1-A5	Method Path (DA)	D:\MassHunter\Data\Users\Anderson\AEiraj>Last\IR-01-117-17_19mg.d\AcqData\DI_04182019.m
Plate Pos.		Target Source Path	D:\MassHunter\PCDL\default.csv
Operator		Result Summary	1 qualified (1 targets)

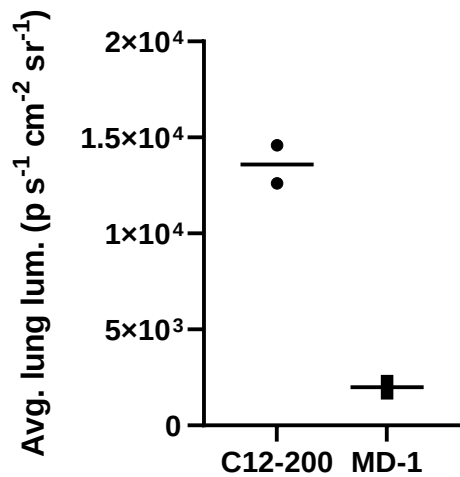
Sample Chromatograms



165
166
167
168
169
170
171
172
173
174
175
176
177



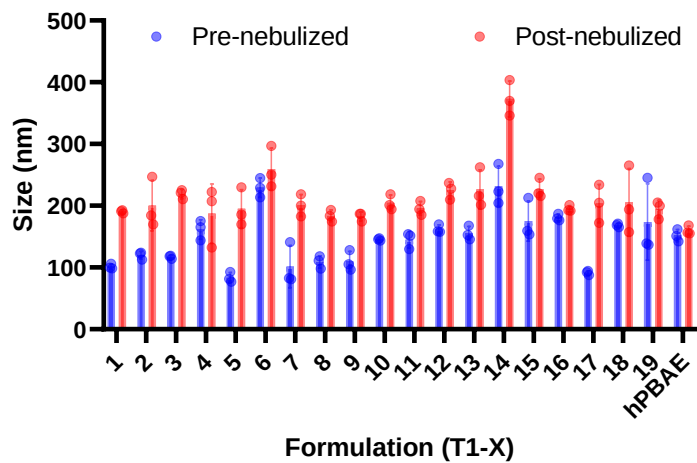
Supplementary Figure 1. Characterizing effects of nebulization on LNPs. Comparison of C12-200 LNP A) transfection efficiency in A549 cells (Average $\pm$ SD, well replicates shown), B) mRNA encapsulation efficiency (Average $\pm$ SD, n = 3 repeated measurements), and C) size as measured by DLS, before and after nebulization. D) Cryo-TEM images of C12-200 LNPs before and after nebulization. Images are representative from two independent experiments. Scale bar = 200nm.



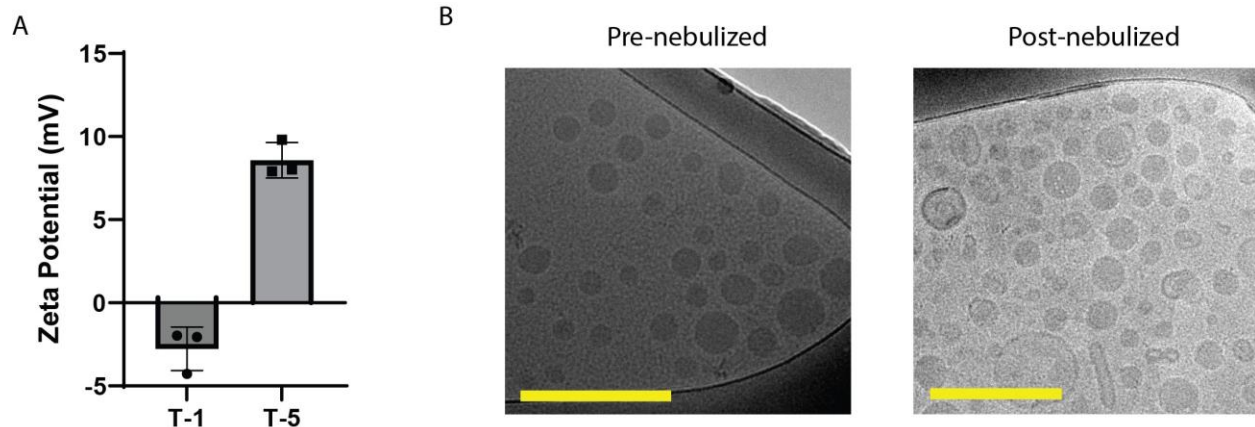
186

187 **Supplementary Figure 2. *In vivo* nebulized mRNA delivery with systemic LNPs.** Comparison of
 188 bioluminescence in lung 6h after nebulized administration of liver-targeting C12-200 LNP
 189 formulation and lung-targeting cKK-E12 LNP formulation delivering 0.5mg FFL mRNA.^{1,2} n = 2
 190 mice per experimental group with mean shown.

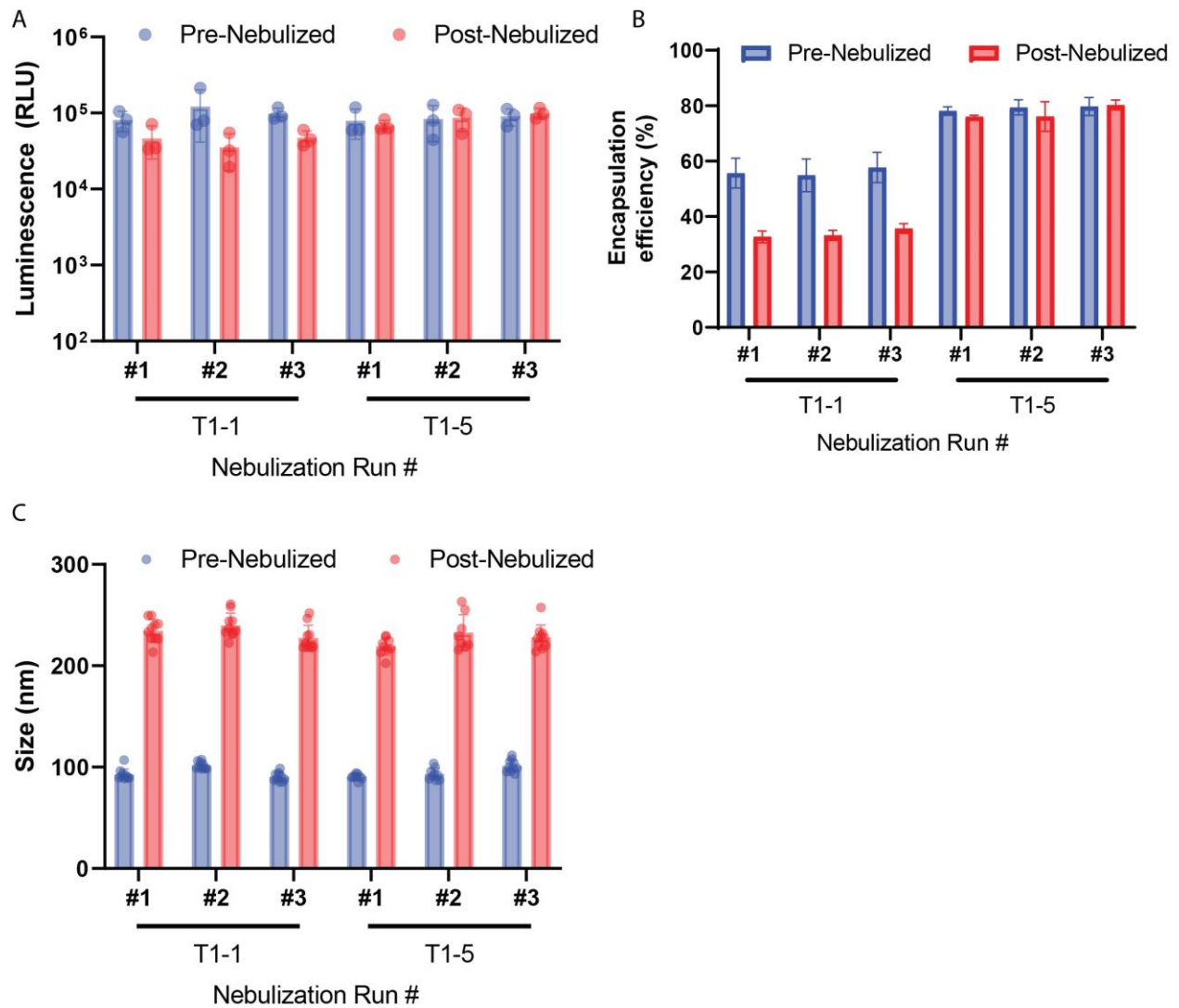
191



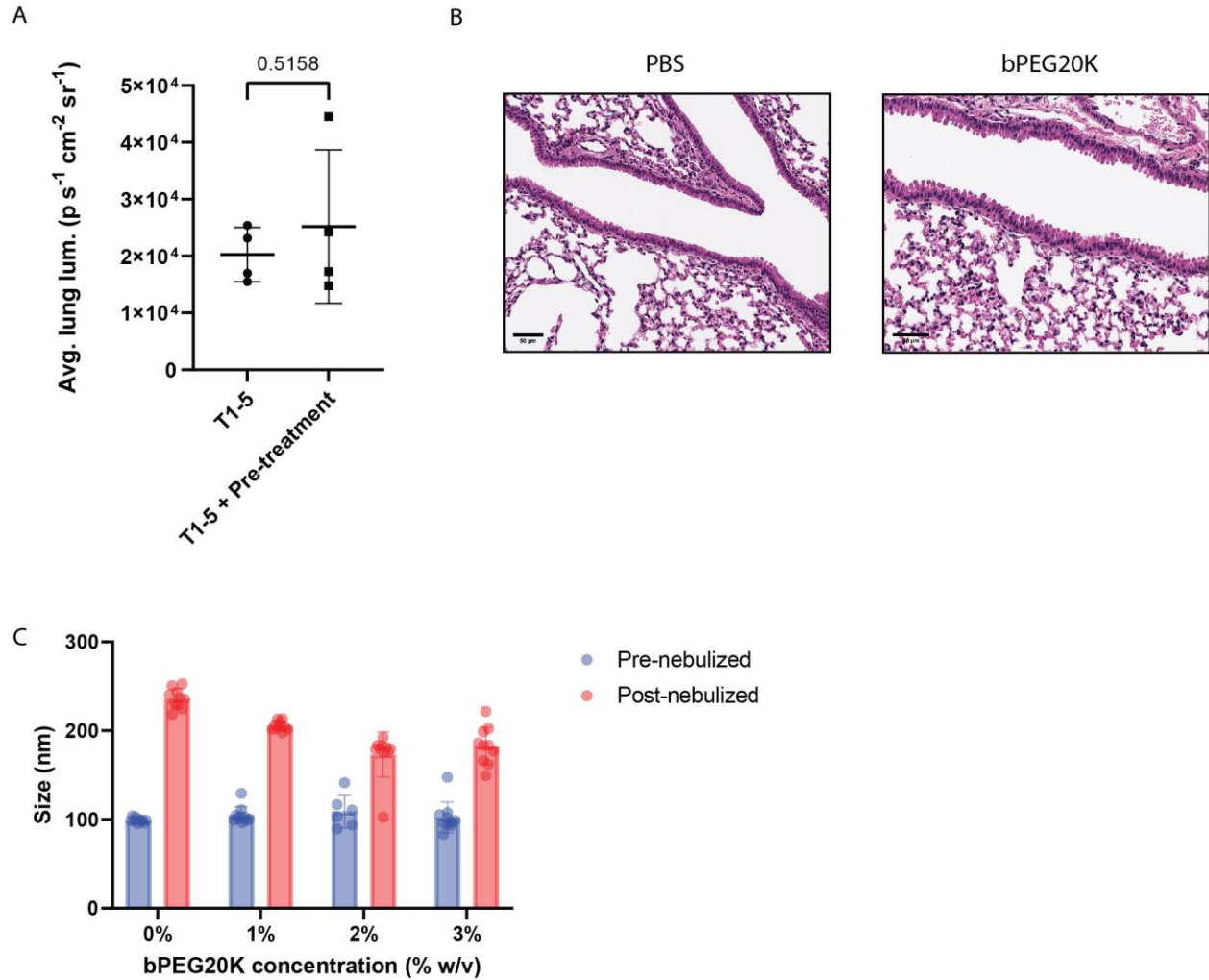
Supplementary Figure 3. DLS of DOE LNPs. DLS measurements of nanoparticle size before and after nebulization for all 19 LNP formulations (T1-1 through T1-19) in the DOE screen and polymeric hPBABE nanoparticles. Average $\pm$ SEM, n=3 repeated measurements.



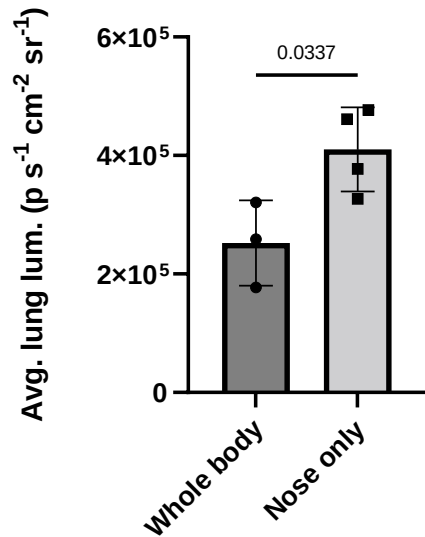
Supplementary Figure 4. Characterization of lead T1-5 LNP from DOE screen. A) Zeta potential of C12-200 T1-1 and T1-5 LNPs. Average $\pm$ SD, n=3 repeated measurements. B) Cryo-TEM images of C12-200 T1-5 LNPs before and after nebulization. Images are representative from two independent experiments. Scale bar = 200nm.



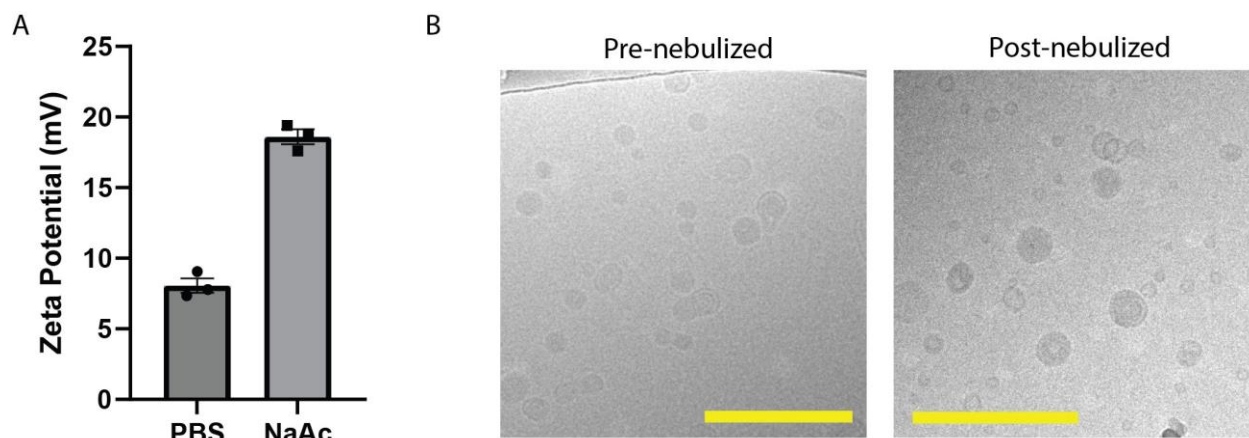
Supplementary Figure 5. Reproducibility of nebulization *in vitro*. Evaluation of A) *in vitro* transfection efficiency in A549 cells (50ng fLuc mRNA/well for pre-nebulized LNPs, post-nebulized LNP dosing was volume matched to pre-nebulized LNP dose, n = 3 wells per treatment group, Average $\pm$ SD), B) encapsulation efficiency (n = 3 repeated measurements, Average $\pm$ SD) and C) LNP size (n = 3 repeated measurements, average $\pm$ SEM) before (blue) and after (red) three independent nebulizations (runs) of T1-1 and T1-5 LNPs to confirm reproducibility of DOE results.



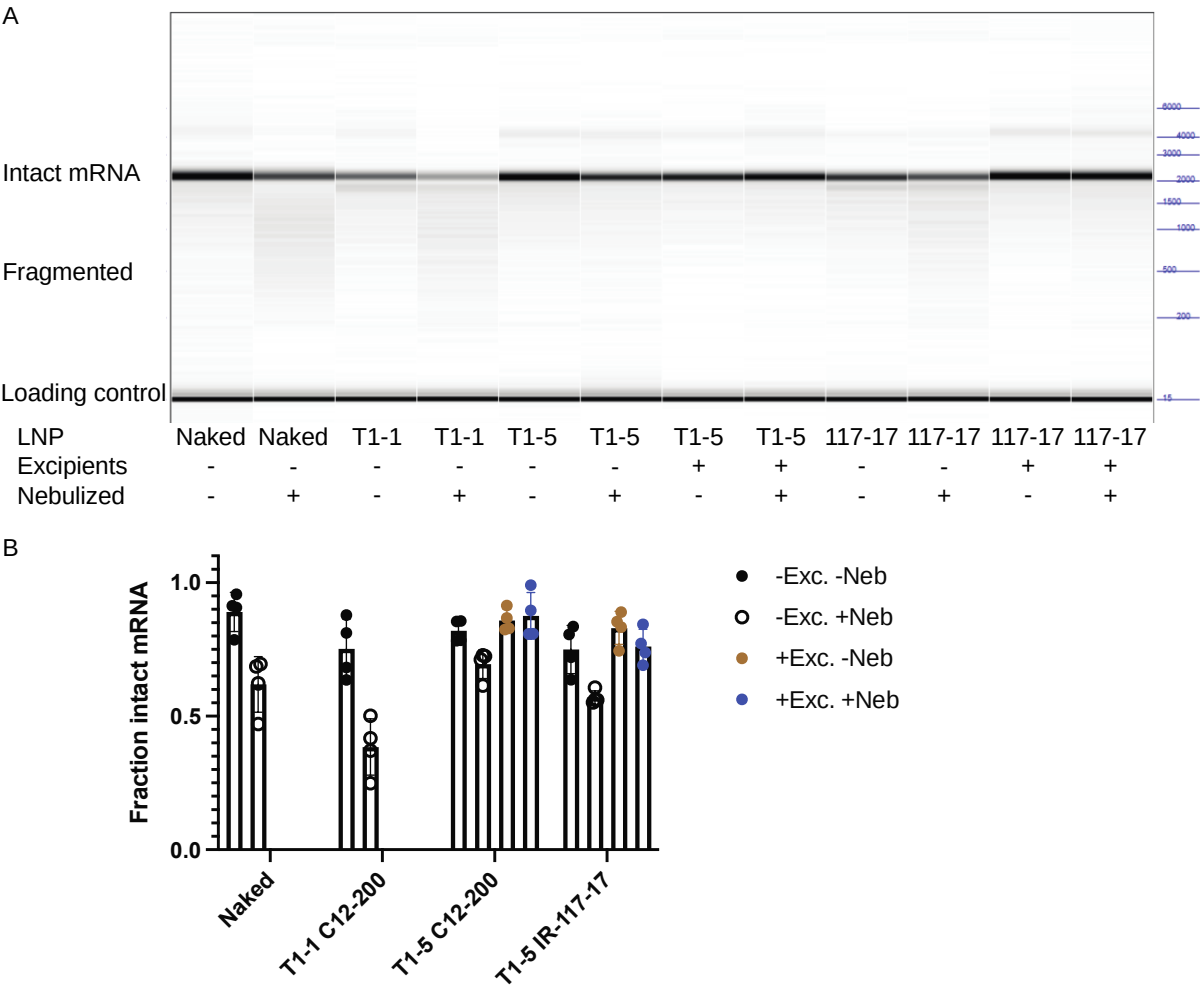
Supplementary Figure 6. Characterization of bPEG20K effects on nebulized LNP formulation and mouse lungs. A) Comparison of luminescence 6h after nebulized delivery of 0.5mg FFL mRNA with T1-5 in lungs either without or with nebulized pre-treatment of 2% bPEG20K solution. For pre-treatment, mice were nebulized 1.5mL of 2% bPEG20K in PBS immediately prior to nebulized delivery of T1-5. Average $\pm$ SD, n = 4 mice per experimental group, two-tailed student's T-test. B) Histology of lungs collected from mice 6h after nebulized administration of either 1.5mL of 2% bPEG20K in PBS or PBS only as a control. Images are representative from one independent experiment with 4 mice in each group. C) DLS measurements of T1-5 LNP size before and after nebulization in the presence of 0%, 1%, 2%, or 3% w/v bPEG20K. Average $\pm$ SEM, n = 10 repeated measurements.



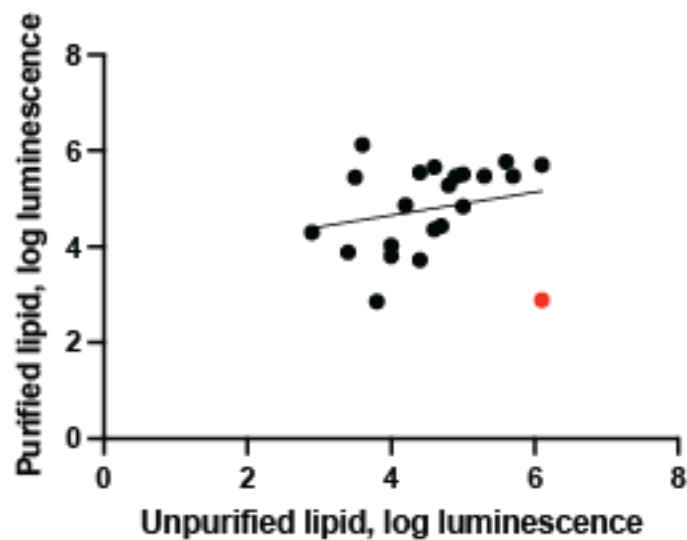
Supplementary Figure 7. Comparison of whole-body exposure and nose-only exposure systems for *in vivo* nebulized mRNA-LNP delivery. Bioluminescence in lungs from mice 6h after nebulized delivery of FFL mRNA using T1-5 LNPs with 2% bPEG20K. Mice were exposed to aerosol either in a whole-body exposure chamber (whole body) or a nose-only exposure system (nose only). Average $\pm$ SD, n=3 mice for whole-body, 4 for nose-only, two tailed student's T-test.



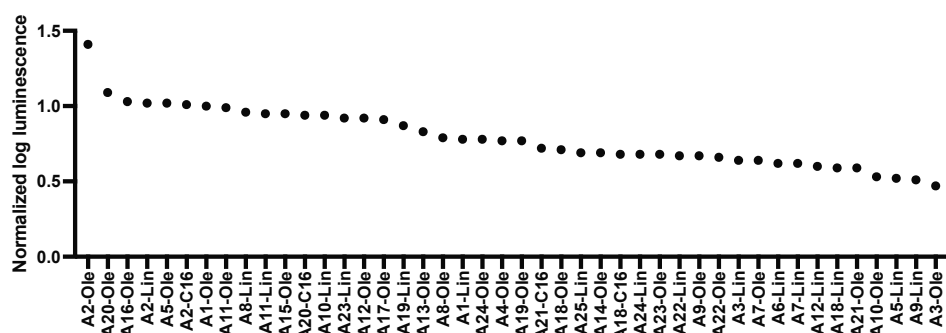
Supplementary Figure 8. Characterization of NaAc dialyzed LNPs. A) Zeta potential of C12-200 T1-5 LNPs following overnight dialysis in either PBS or sodium acetate (NaAc) buffer with 2% w/v bPEG20K excipient added to the LNP solution prior to measurement. Average $\pm$ SD, $n = 3$ repeated measurements. B) Cryo-TEM images of LNPs in NaAc buffer with 2% bPEG20K before and after nebulization. Images are representative of two independent experiments. Scale bar = 200nm



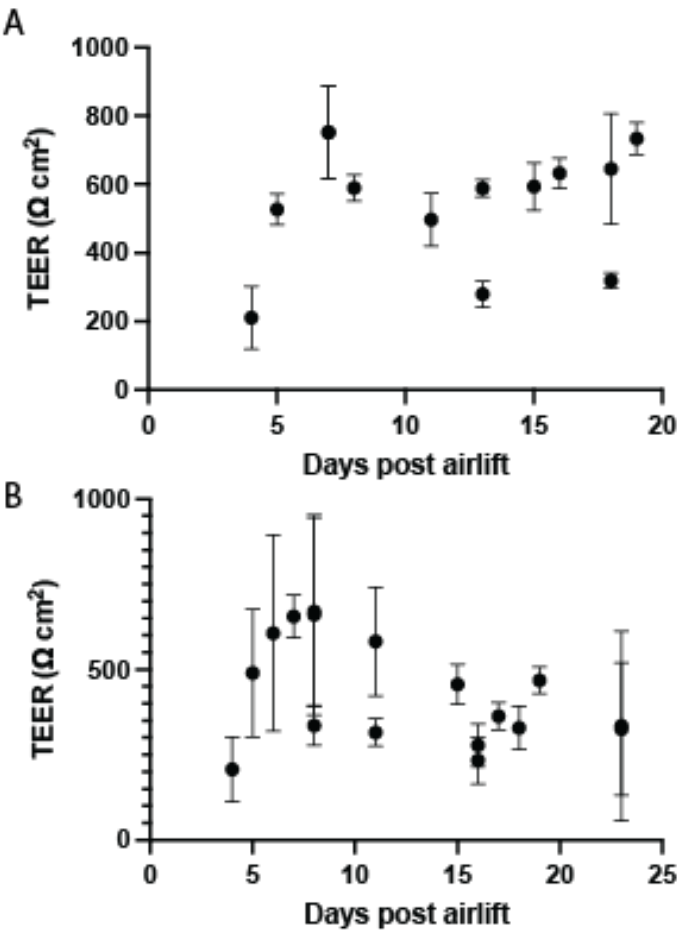
Supplementary Figure 9. Effects of nebulization on mRNA integrity. A) Representative (out of n=4 total experiments performed as plotted in (B)) image from Fragment Analyzer trace of mRNA extracted from LNPs pre- or post-nebulization. Lower molecular weight smears below main band indicates fragmentation. B) Fragment Analyzer traces quantified by the fraction of RNA in the “intact mRNA” band versus intact mRNA + fragmented RNA. Plots indicate average ± SEM (n=4). IR-117-17 described below.



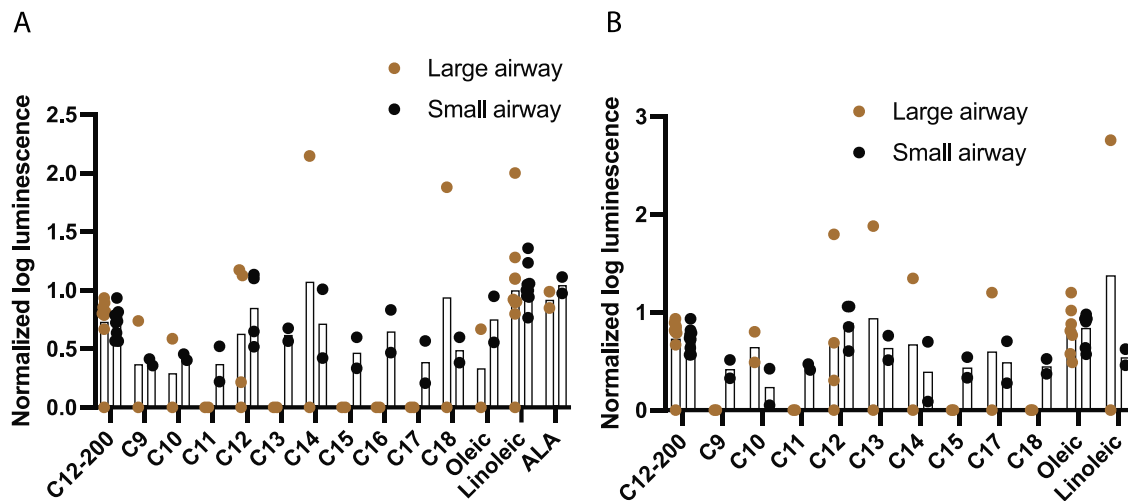
Supplementary Figure 10. Comparison between purified and unpurified ionizable lipids. Correlation between FFL delivery of T1-5 LNPs generated from purified and unpurified lipids. Correlation is not significant ($p=0.66$) and removal of outlier (red) does not yield a significant correlation ($p=0.15$); correlation p values calculated using F test.



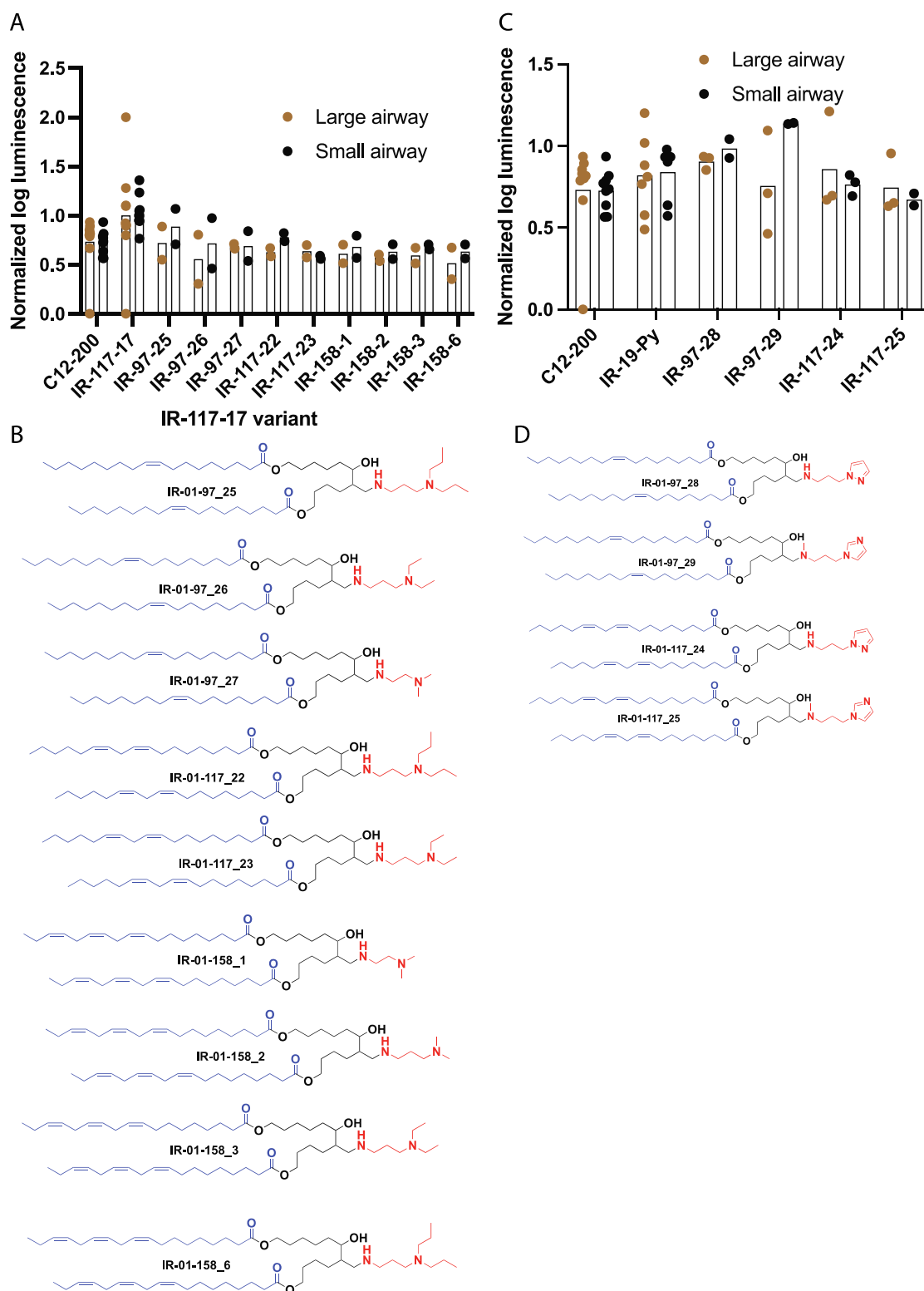
Supplementary Figure 11. A549 cell screening of novel ionizable lipids. FFL mRNA delivery screen to A549 cells using the T1-5 formulation, normalized in log space to C12-200 delivery, using handmixed non-nebulized LNPs (20ng/well).



Supplementary Figure 12. Trans-epithelial electrical resistance (TEER) of air-liquid interface (ALI) cultures. A) TEER of ALI cultures generated from primary large airway cells (n>3 per measurement, ALI cultures from 3 different donors included). B) TEER of ALI cultures generated from primary small airway cells (n>3 per measurement, ALI cultures from 2 different donors included). Average, +/- SD.

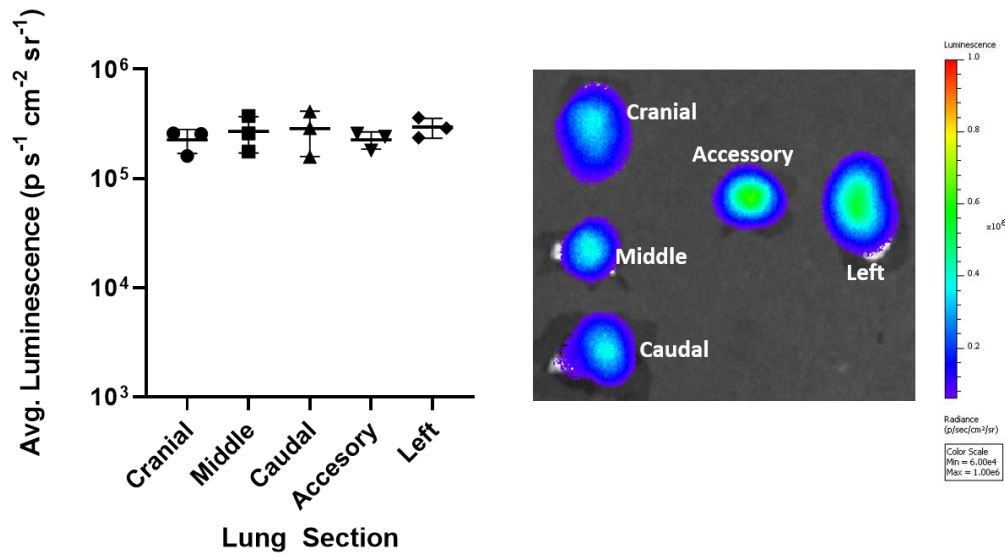


Supplementary Figure 13. ALI culture screen of lead lipid tail variants. Screen of tail variants of (A) IR-117-17 and (B) IR-19-Py (n=2 per airway cell line type, i.e. 4 wells total/lipid, 500ng mRNA/well). Each value for a lipid and cell type is normalized to IR-117-17's delivery to that cell type. Testing performed using handmixed non-nebulized LNPs.

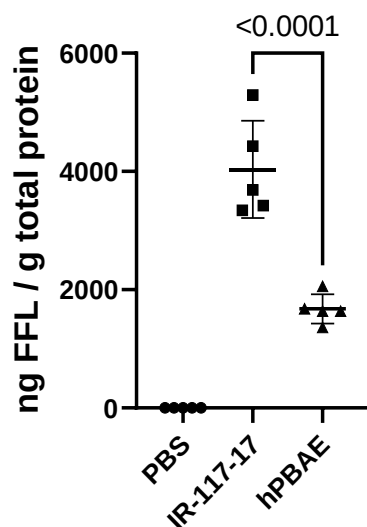


Supplementary Figure 14. ALI culture screen of lead lipid headgroup variants. Screen of headgroup variants (plus tailgroup variants of the headgroup variants themselves) of (A) IR-117-17 (structures shown in (B)) and (C) IR-19-Py (structures shown in (D)) (n=2 per airway cell line type). Each value for a lipid and cell type is normalized to IR-117-17's delivery to that cell type.

277 C12-200 data is shared for this figure and Supplementary Figure 12. Testing performed using
278 handmixed non-nebulized LNPs.
279

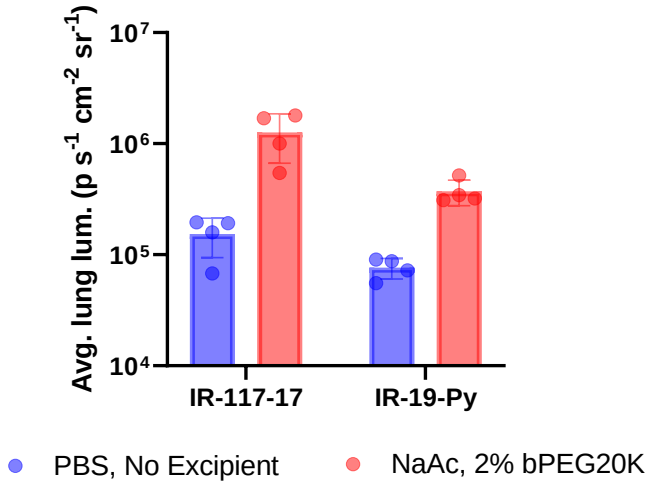


Supplementary Figure 15. Biodistribution of luciferase in lung lobes. Uniform distribution of luminescence across all 5 lobes of the lung (left), with representative image of distribution (right), following nebulized delivery of 1mg FFL mRNA with IR-117-17 LNPs in NaAc buffer + 2% w/v bPEG20K using a nose-only delivery chamber. Average $\pm$ SD, n = 3 mice.



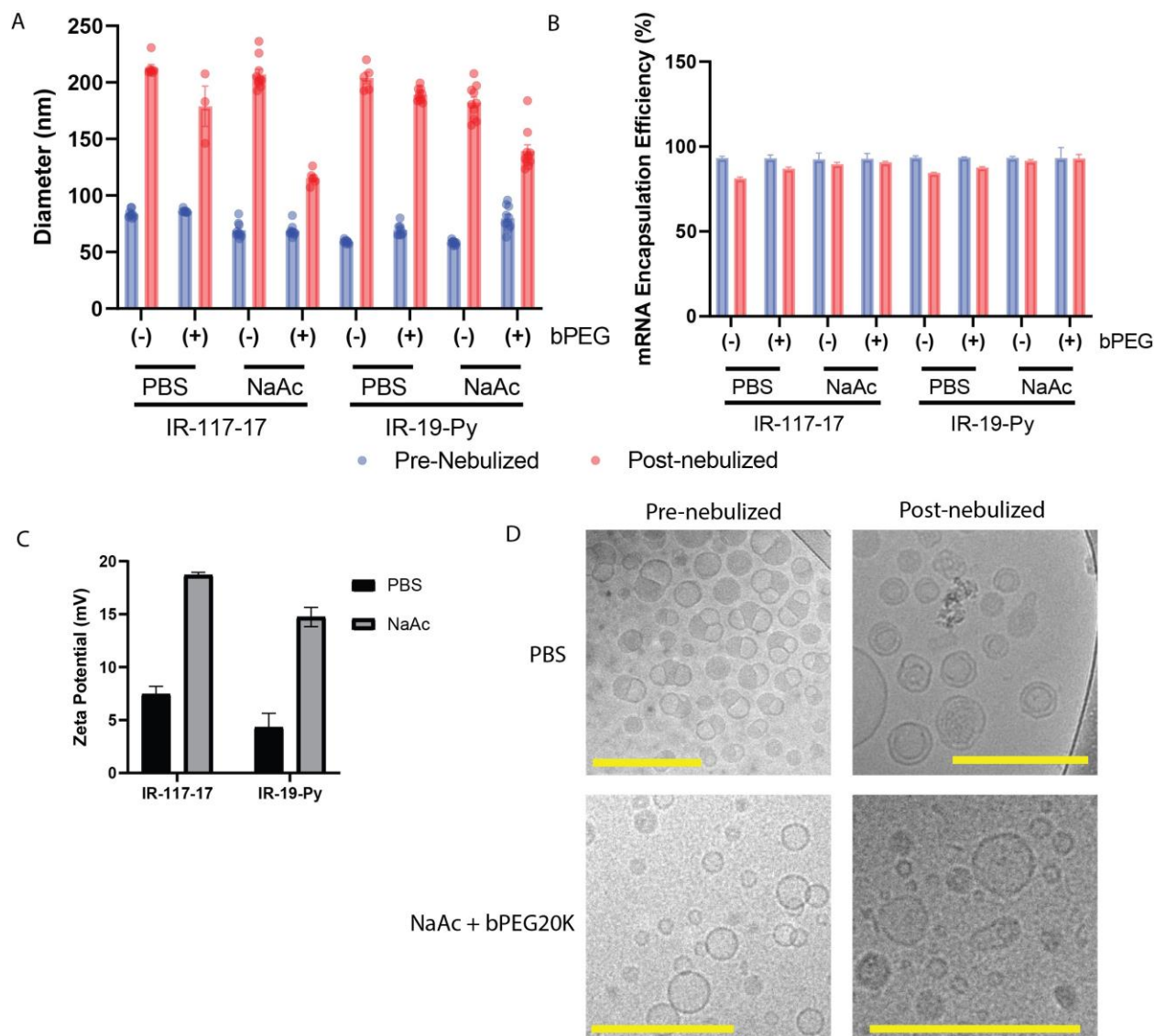
Supplementary Figure 16. Quantification of FFL protein expression in homogenized lung tissues. 1mg of FFL mRNA was nebulized to the lung with either IR-117-17 LNPs in NaAc buffer + 2% w/v bPEG20K, hpBAE, or PBS as control. Lungs were collected at 6h post-nebulization for IR-117-17 and 24h post-nebulization for hpBAE. Two-tailed Student's *t*-test, Average ± SD, n = 5 mice per experimental group.

296
297

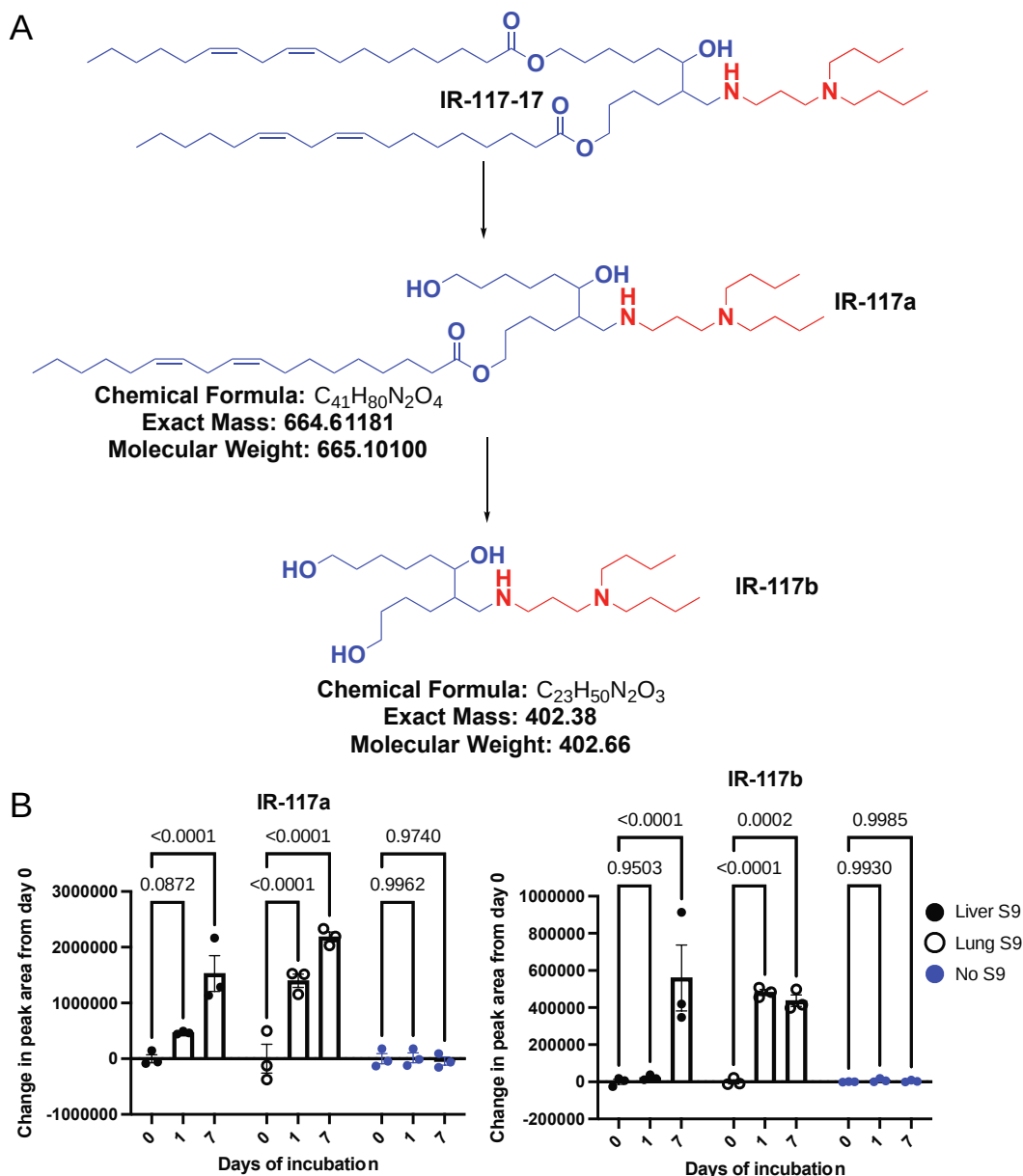


298
299
300
301
302
303
304

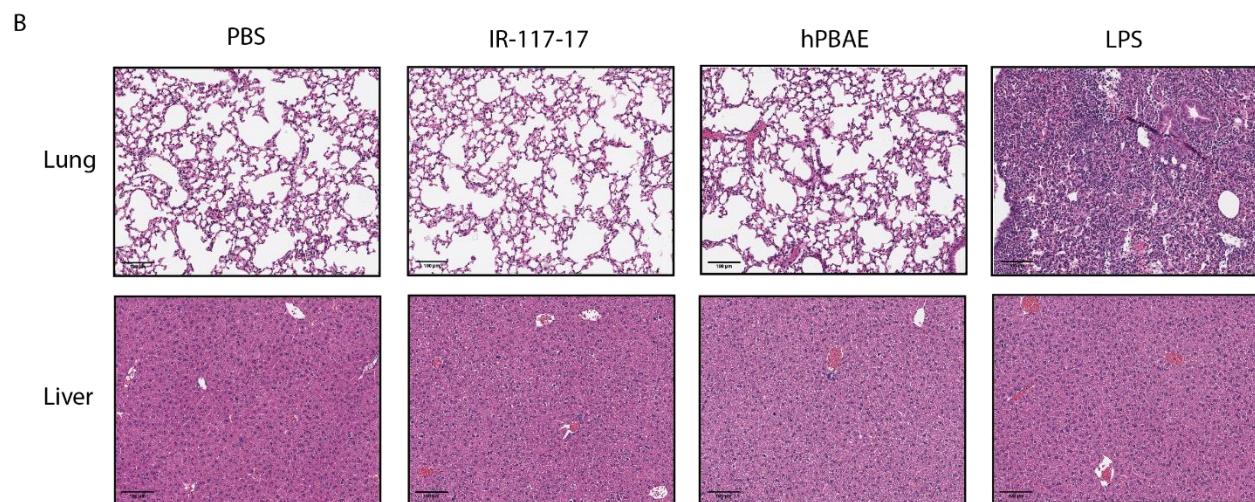
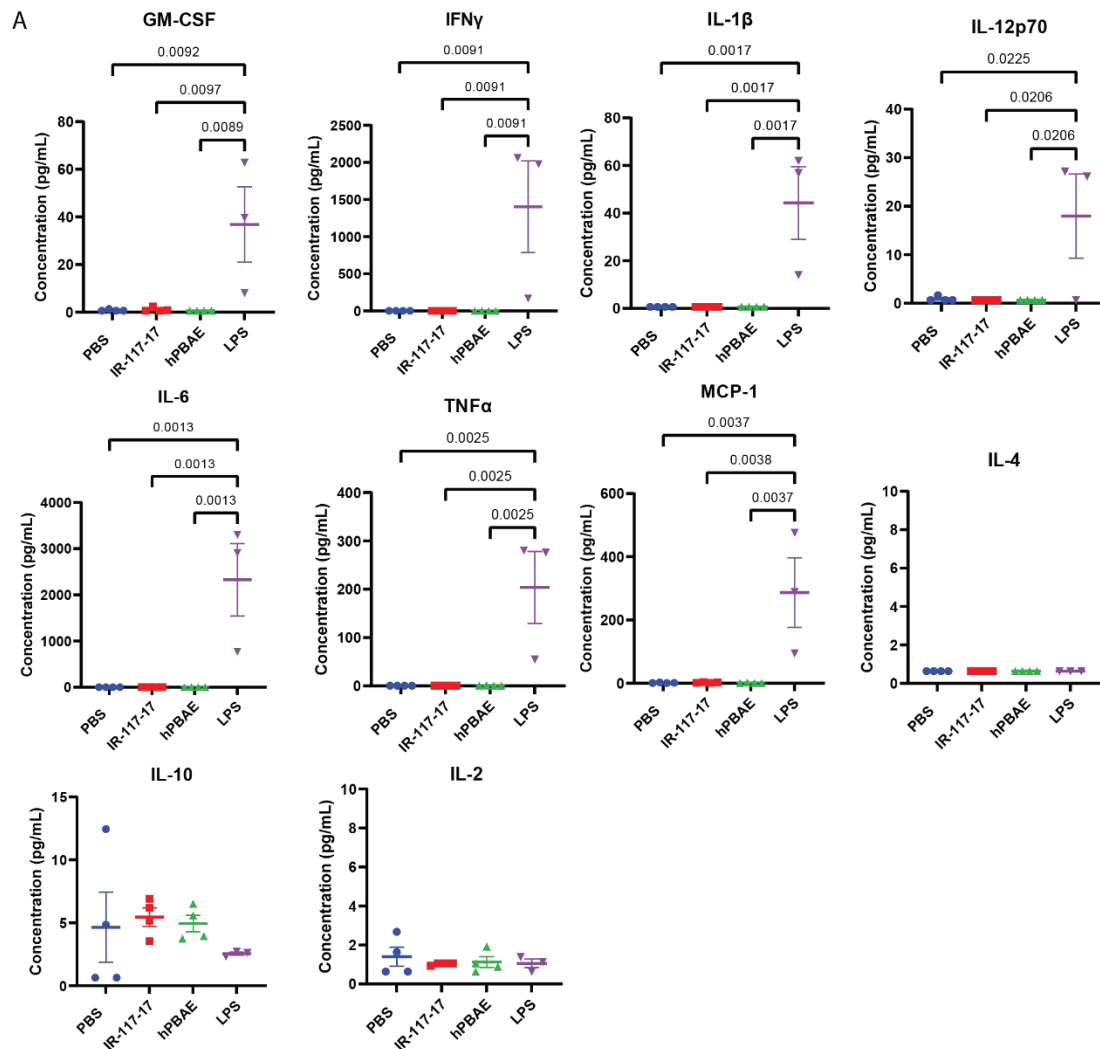
Supplementary Figure 17. Optimized nebulization formulation for reductive amination lipids. Comparison of *in vivo* nebulized delivery of 1mg of FFL mRNA with LNPs comprising the biodegradable lipids IR-117-17 or IR-19-Py. Addition of NaAc buffer and bPEG20K improves nebulized delivery of both LNPs compared to PBS. Average ± SD, n = 4 mice per experimental group.



Supplementary Figure 18. Physicochemical characterization of optimized nebulization formulation for IR-117-17 and IR-19-Py LNPs. A) DLS measurements of LNP size before and after nebulization in the presence or absence of 2% w/v bPEG20K and in either PBS or NaAc buffer. Average $\pm$ SEM, n=10 repeated measurements. B) FFL mRNA encapsulation efficiency of IR-117-17 and IR-19-Py LNPs before and after nebulization in the presence or absence of 2% w/v bPEG20K and in either PBS or NaAc buffer. Average $\pm$ SEM, n=3 repeated measurements. C) Zeta potential of IR-117-17 LNPs dialyzed overnight in either PBS or NaAc buffer. 2% w/v bPEG20K was added to the LNP solution prior to measurement. Average $\pm$ SEM, n = 3 repeated measurements. D) Cryo-TEM images of IR-117-17 LNPs before and after nebulization in either PBS buffer with no excipient or NaAc buffer with 2% w/v bPEG20K (NaAc + bPEG20K). Scale bar = 200nm.



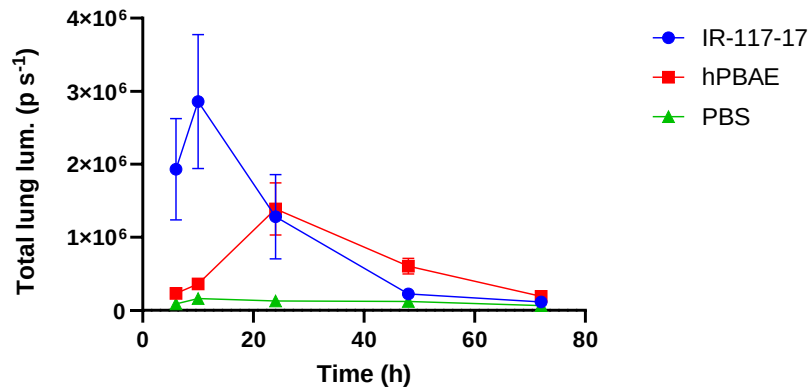
Supplementary Figure 19. Degradation analysis of IR-117-17. A) Proposed degradation pathway for IR-117-17. B) Change in area of LCMS peak corresponding to each degradation product after 0, 1, or 7 days of incubation with 2mg/mL human liver S9 fraction, lung S9 fraction or PBS control (no S9). Two-way ANOVA with Dunnett's multiple comparisons test, mean $\pm$ SD (n = 3).



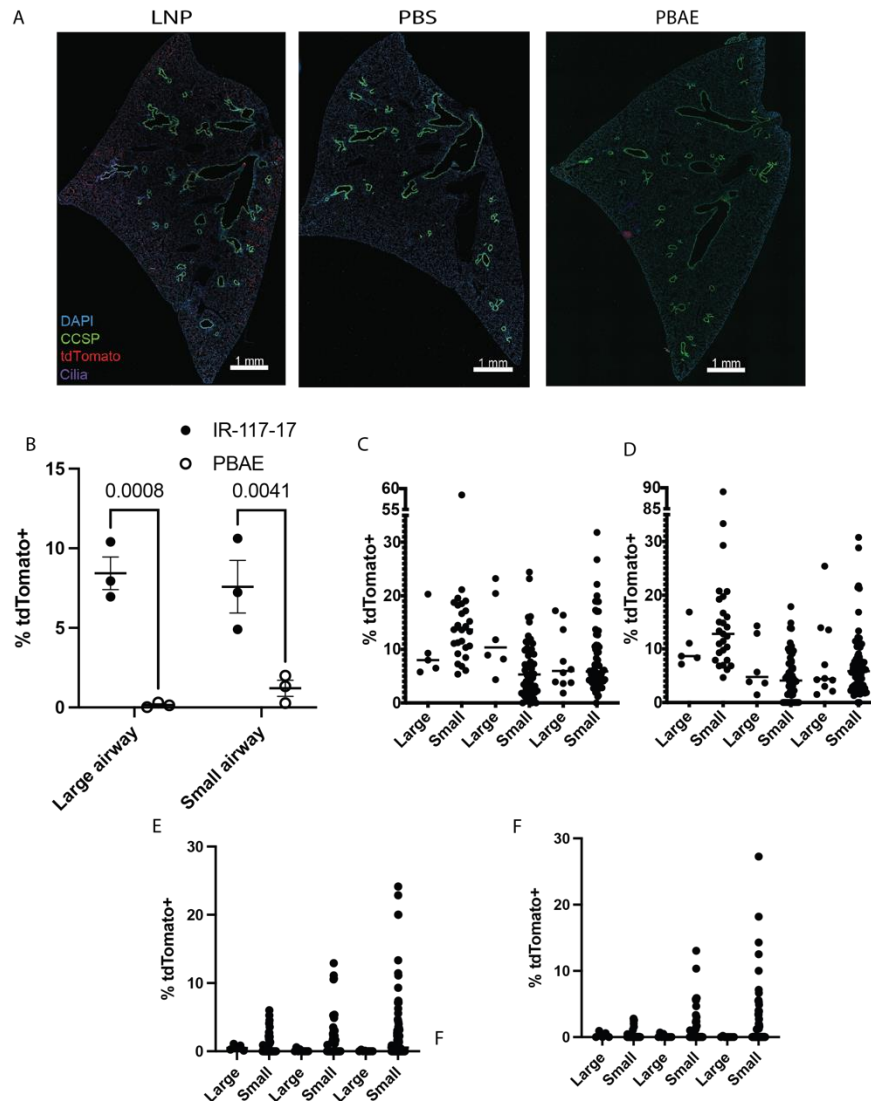
Supplementary Figure 20. Safety and toxicity of nebulized LNPs. A) Concentrations of pro-inflammatory cytokines in BALF collected from mice 24h after nebulized administration of 1mg of FFL mRNA with either IR-117-17 or hPBAE. Average $\pm$ SEM, n = 3 mice for LPS, n = 4 for all

330 other groups, one-way ANOVA with post-hoc Tukey test. B) Histology of lungs and livers
331 collected from mice 24h after nebulized administration of 1mg of FFL mRNA. Normal tissue
332 histology is demonstrated in all samples except LPS treated mice exhibited substantial immune
333 cell infiltration in the lung. PBS was nebulized as a negative control while LPS was intranasally
334 administered as positive control for inflammation. Images are representative from one
335 independent experiment with n=3 mice for LPS, 4 for all other groups.

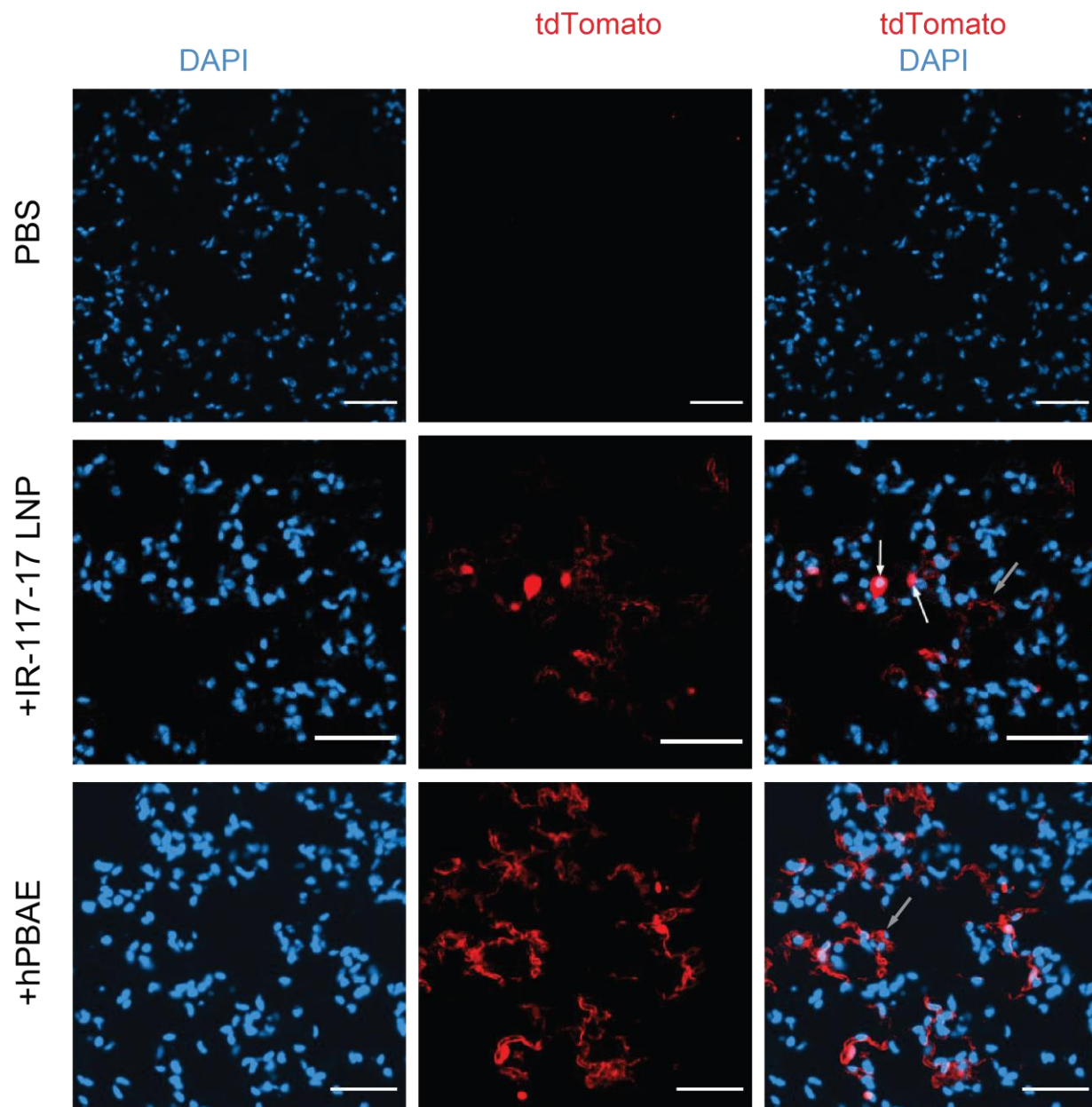
336



Supplementary Figure 21. Kinetics of luciferase expression in the lung of *Scnn1b*-Tg mice. 1mg of FFL mRNA was nebulized to *Scnn1b*-Tg mice with either IR-117-17 or hPBAE at time t = 0h. PBS was nebulized as a control. Average $\pm$ SD, n = 3 mice for PBS, 5 mice for other groups.



Supplementary Figure 22. Ai14 mouse model transfection analysis. A) Representative whole slides that were quantified for tdTomato delivery. Images are representative from one independent experiment with 3 mice per treatment group and two slide per mouse. B) Quantification of transfected club (CCSP+) cells in Ai14 mouse large and small airways by IR-117-17 LNPs and PBAE (n=3 mice, average $\pm$ SEM, 2-way ANOVA). C-F) Quantification of transfected cells in individual airways for (C) all airway cells transfected with nebulized IR-117-17 LNP, (D) club cells (CCSP+) with LNP, (E) all airway cells transfected with nebulized PBAE, (F) club cells with PBAE.



Supplementary Figure 23. Transfection of alveolar spaces. Sample images of alveolar spaces from mice treated with nebulized PBS, 9mg nebulized Cre mRNA in IR-117-17 LNPs, and 9mg nebulized Cre mRNA in hPBAE. White arrows indicate transfected immune cells, gray arrows indicate selected transfected alveolar cells. Scale bar = 50 μ m.

Supplementary Table 1. Formulation parameters for DOE study in Figure 1. All formulations included 2.5 mol% C₁₄-PEG_{2K} and were formulated at 10:1 ionizable lipid:mRNA weight ratio.

Formulation ID	Ionizable lipid	Phospholipid	Mole percentage (%)		
			Ionizable lipid	Phospholipid	Cholesterol
T1-1	C12-200	DOPE	35	16	46.5
T1-2	C12-200	DOPE	25	16	56.5
T1-3	C12-200	DOPE	45	16	36.5
T1-4	C12-200	DOTAP	25	16	56.5
T1-5	C12-200	DOTAP	35	28	34.5
T1-6	C12-200	DOTAP	45	4	48.5
T1-7	C12-200	DSPC	25	28	44.5
T1-8	C12-200	DSPC	37	4	56.5
T1-9	C12-200	DSPC	45	16	36.5
T1-10	cKK-E12	DOPE	25	16	56.5
T1-11	cKK-E12	DOPE	33	28	36.5
T1-12	cKK-E12	DOPE	35	16	46.5
T1-13	cKK-E12	DOPE	45	4	48.5
T1-14	cKK-E12	DOTAP	25	28	44.5
T1-15	cKK-E12	DOTAP	37	4	56.5
T1-16	cKK-E12	DOTAP	45	16	36.5
T1-17	cKK-E12	DSPC	25	16	56.5
T1-18	cKK-E12	DSPC	33	28	36.5
T1-19	cKK-E12	DSPC	45	4	48.5